→ Outlook → Temperature → Humidity → Windy → Rainy

Date: (1)

OL	T	H	W	R
Sunny	hot	high	weak	NO
Sunny	mild	high	Strong	NO
Cloudy	hot	high	Weak	yes
Rainy	mild	high	weak	yes
Rainy	cool	normal	weak	yes
Rainy	Cool	normal	Strong	NO
Cloudy	mild	normal	Strong	yes
Sunny	Cool	normal	weak	no

Total record => 8

Yes => 4

NO => 4

formula :> $H(S) = i \Sigma p i \log_2 \log$

Date:

Calculate the entropy of Raing

Yes = 4 / NO = 4

$$\Rightarrow -\frac{Yes}{5}\log_2\frac{yes}{8} - \frac{NO}{8}\log_2\frac{NO}{8}$$

$$-\frac{4}{5}\log(2)\frac{4}{8} - \frac{4}{8}\log(2)\frac{4}{8}$$

$$-0.07525 + 0.07525$$

$$\Rightarrow -0.1505$$

Calculate the Outlook attribute information gain & weighted entropy

outlook	Total	Yes	no	entropy
Sunny	3	0	3	0.000
Cloudy	2	2	0	0.000
Rainy	3	2	1	0.1671

Sunny

$$-0\log(2)0 - \frac{1}{3}\log(2)\frac{1}{3}$$

$$-\frac{1}{3}\log(0) \div \log(2) - \frac{1}{3}\log(\frac{1}{3}) \div \log(2)$$

$$0 + 0$$

$$\Rightarrow 0$$

$$(Cloudy) -\frac{2}{2}\log(2)\frac{2}{2} - \frac{0}{2}\log(2)\frac{0}{2}$$

$$\Rightarrow -1\log(2)1 - 0$$

$$0 \Rightarrow 0.00$$

Date:

Rainy

$$-\frac{2}{3}\log_2\frac{2}{3} - \frac{1}{3}\log_2\frac{1}{3}$$

$0.1337 + 0.0334$

$\Rightarrow 0.1671$

weighted entropy

$\frac{3}{8}(0.000) = 0$

$\frac{2}{8}(0.000) = 0$

$\frac{3}{8}(0.1671) = 0.0626$

$-0.1505 - 0.0626$

$\Rightarrow \boxed{0.2131}$

Calculate the entropy of Temprature Info gain & weighted entropy

Temprature	Total	yes	no	entropy
mild	3	2	1	0.1671
hot	2	1	1	0.1504
cold	3	1	2	0.1671

Date:

mild $= -\frac{2}{3}\log(2)\frac{2}{3} - \frac{1}{3}\log(2)\frac{1}{3}$

$\Rightarrow 0.1337 + 0.0334$

0.1671

hot $= -\frac{1}{2}\log(2)\frac{1}{2} - \frac{1}{2}\log(2)\frac{1}{2}$

$0.0752 + 0.0752$

HOT $\Rightarrow 0.1504$

Cold $\Rightarrow -\frac{1}{3}\log(2)\frac{1}{3} - \frac{2}{3}\log(2)\frac{2}{3}$

$0.0334 + 0.1337$

$\Rightarrow 0.1671$

Weighted

$\frac{3}{8}(0.1671) = 0.0626$

$\frac{2}{8}(0.1504) = 0.0376$

$\frac{3}{8}(0.1671) = 0.0626$

$\Rightarrow 0.1628$